DO NOW — RECALL

Arrays, records and tuples

Name: _____ Class: _____ Date: _____

1. Why must exponents match before adding binary mantissas?

2. What does passing an int by value mean?

3. What is values[1] for int[] values={4,6,9}?

4. a={1,2,3}; a[0]=a[1]+a[2]. Find a.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The mantissa bit positions must have matching place values.

2. Model answer

The method receives a copy of the integer.

3. Model answer

6.

4. Model answer

[5,2,3].

Vocabulary — know these terms

array — A fixed-length indexed collection of values of one element type in C#.

index — Zero-based position of an array element.

record — A structure that groups related named fields or properties.

tuple — A compact ordered grouping of multiple values.

assignment — Store a new value at a variable or array location.

NEW LEARNING

Read, annotate and explain

Zero-based indexing

scores[0] is 4 and scores[2] is 2.

Read before writing

The right-hand expression is 4 + 2 = 6.

Replace the selected value

Assign 6 at index 1; the array now contains 4, 6, 2.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace marks = {5,7,3}; marks[2] = marks[0] + marks[1].

Predict

marks[0] + marks[1] means 5 + 7.

Confirm and explain

12 replaces the value at index 2: [5,7,12].

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give the final array after int[] a={8,4,2}; a[1]=a[0]-a[2];

Question 2

When would you choose a record rather than a simple array for a pupil's name and mark?

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain a suitable use of a named tuple in C# and one limitation.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An array with three values has a last index of 3.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.